

Exact parameter identification in PET pharmacokinetic modeling using the irreversible two tissue compartment model

Group members:

2105013 - Tasnimul Islam Mahi

2105022 - Mafruha Islam Aishi

2105024 - Ridhwanul Islam

2105025 - Ashrafur Rahman

2105026 - Tanvir Liaquat Uday

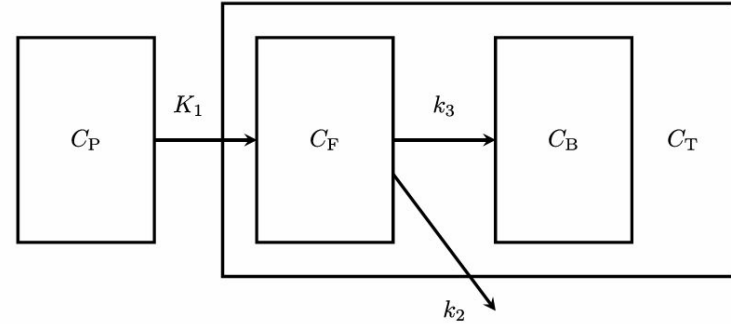
Subsection : A1

Base Paper Overview

- Dynamic PET measures how a radioactive tracer moves through blood and tissue over time.
- The paper uses an irreversible two-tissue compartment model to estimate three kinetic parameters for each brain region:
 - K_1 : tracer transfer from blood into tissue
 - k_2 : tracer returning from tissue to blood
 - k_3 : irreversible trapping or metabolism
- Normally, estimation requires repeated arterial blood sampling.
- The paper mathematically shows that most parameters can potentially be recovered using PET time-activity curves from multiple regions, without complete arterial blood measurements.
- It uses Tikhonov regularization and an iteratively regularized Gauss–Newton method to handle noisy data.

Problem Statement and Project Scope

Problem: Estimating K_1 , k_2 , and k_3 is a nonlinear inverse problem. PET measurements contain noise, so ordinary parameter fitting may be unstable or inaccurate.



Proposed scope

- Generate synthetic arterial-input and tissue time-activity curves.
- Add controllable random noise to simulate PET measurement uncertainty.
- Estimate K_1 , k_2 , and k_3 using numerical optimization.
- Compare results under different noise levels and initialization errors.
- Investigate whether regularization improves estimation stability.
- Evaluate performance using parameter error, curve-fitting error, convergence, and computation time.

Expected Methodology

- Model arterial tracer concentration using exponential functions.
- Generate synthetic tissue time-activity curves using the two-tissue compartment model.
- Add Gaussian or Poisson noise to simulate PET measurements.
- Estimate K_1 , k_2 , and k_3 using regularized Gauss–Newton optimization.
- Solve linearized systems using LU or QR decomposition.
- Evaluate results using relative error, RMSE, convergence, and noise sensitivity.
- Plot true, noisy, and reconstructed curves.

Course Topics Applied

- Random-number generation and Monte Carlo simulation
- Newton/gradient-based optimization
- Constrained least-squares regression
- LU/QR decomposition
- Numerical integration and interpolation
- Error analysis and visualization

Thank youwu !!!